Cloud Counselage

GIFT A CAREER FOUNDATION

GLOBAL PROFESSIONAL INTERNSHIP (GPI)

PROJECT REPORT

<Data Analytics>

A Data-Driven Study on Student Academic Performance, Skills, and Expected Salary

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1. PROJECT DETAILS

Project Name	**Student Career Insights using Data Analysis**		
Project Sponsor			
Project Manager			
Start Date	**16-01-2026**	**21-02-2026**	

2. SUMMARY

This project focuses on examining how various factors such as a student's financial background, academic performance, technical skills, leadership abilities, and experience influence their expected salary.

The analysis was conducted using Python-based data analysis and visualization techniques in Google Colab.
The motivation behind this project stems from the common belief among students that economic background plays a dominant role in career success. However, such assumptions are often not supported by data.

This project aims to validate or challenge these beliefs through a structured, data-driven approach.

By performing exploratory data analysis (EDA) and creating visual dashboards, the project uncovers meaningful relationships between different variables. The results highlight that consistent academic performance, relevant technical skills (such as Python), and leadership exposure contribute more significantly to salary expectations than financial background alone.

These findings can help students make informed career decisions, assist educational institutions in refining curriculum strategies, and support career advisors in guiding students effectively.

3. INTRODUCTION

a. Background

With the growing number of students applying for internships and job roles, the competition in recruitment processes has become increasingly intense. Recruiters typically spend only a few minutes reviewing each resume, leading to a high rejection rate at the initial screening stage.

During the initial requirement analysis, it was observed that students often lack clarity about which factors truly impact employability and salary expectations. While many believe that family income plays a major role, other variables such as GPA, technical expertise, and leadership experience may have a stronger influence

To address this uncertainty, this project uses structured student data and applies Python-based analysis in Google Colab to identify patterns and relationships between these factors.

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b. Stakeholders

- **Project Mentors:** Provide guidance and technical direction
- **Academic Coordinators:** Monitor project progress and evaluation
- **Students:** Benefit from insights for career planning
- **Educational Institutions:** Use findings for curriculum improvement

c. Objectives

The key objectives of this project are:

- To examine the relationship between GPA, family income, Python experience, leadership skills, and expected salary
- To analyze student distribution based on city, college, and graduation year
- To identify trends in participation in academic and extracurricular events
- To present findings through clear and interactive visualizations

4. METHODOLOGY

These conventions are all about the positions of line breaks, how many characters should go on a line, and everything in between.

d. Considerations & Assumption

The following assumptions were made during the project:

- The dataset used is accurate and reliable
- Salary expectations provided by students reflect realistic estimates
- The scope of the project is limited to descriptive and visual analysis
- Missing or inconsistent data does not significantly affect overall conclusions

e. Approach

A systematic approach based on Exploratory Data Analysis (EDA) was followed. The project was implemented using Python in Google Colab, leveraging libraries such as Pandas, Matplotlib, and Seaborn. The workflow included:

1. Understanding the dataset and problem statement
2. Cleaning and preprocessing the data
3. Performing exploratory data analysis
4. Creating visualizations to identify trends
5. Interpreting results and deriving insights

This approach ensured that complex data relationships were presented in a simple and understandable format.

f. Activities

The following steps were carried out during the project:

- Requirement analysis and objective definition
- Data collection and initial exploration
- Data cleaning and preprocessing using Python
- Visualization of data using graphs and charts
- Analysis of relationships between variables
- Creation of dashboards for better interpretation
- Documentation of results and conclusions

5. TARGETTED V/S ACHIEVED OUTPUT

The initial goal of the project was to analyze and visually represent how different factors influence student salary expectations and employability.

The final outcome successfully met these objectives. Key relationships between variables such as GPA, technical skills, and leadership experience were effectively identified and visualized using Python in Google Colab.

Although predictive modeling was not implemented due to dataset limitations, the descriptive and visual analysis provided strong and meaningful insights. This limitation did not affect the overall success of the project but highlighted opportunities for future enhancement.

6. CONCLUSION

The analysis clearly indicates that academic performance, technical expertise (especially in Python), and leadership experience play a more significant role in determining expected salary than economic background alone.
These insights emphasize the importance of skill development and consistent academic effort for students aiming to improve their career

prospects.
For institutions and mentors, the findings provide a data-driven foundation for improving training programs and guiding students more effectively.
Future improvements may include:

- Using larger and more diverse datasets
- Implementing machine learning models for prediction
- Incorporating real-time job market data

7. APPENDICES

g. Appendix A – Title

Component	Description
Student Distribution	City-wise and college-wise student distribution
GPA Analysis	Average GPA by city, college, and leadership skills
Salary Analysis	Expected salary vs GPA, experience, and family income
Event Participation	Field-wise and event-wise participation analysis
Skill Analysis	Impact of Python experience and leadership skills